Math 241, F14 | Name: Answer

Quiz 4 | SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

Problem 1. (4 points) Compute

$$\lim_{t\to+\infty} < \arctan t,\ e^{-2t},\ \frac{\ln t}{t} >$$

$$= < \lim_{t\to+\infty} \arctan t, \lim_{t\to+\infty} e^{-2t}, \lim_{t\to+\infty} \frac{\ln t}{t} >$$

$$= < \frac{\pi}{2}, 0, 0 >$$

Problem 2. (6 points) Let $\mathbf{r}(t) = e^t\mathbf{i} + \sqrt{2}t\mathbf{j} + e^{-t}\mathbf{k}$ that defines a space curve.

(a) Find the unit tangent vector $\mathbf{T}(t)$ of the curve at the point with $t = 0$.

$$r'(t) = e^t i + \sqrt{2} j - e^{-t} k \Rightarrow r'(0) = < 1, \sqrt{2}, -1 >$$

$$|r'(0)| = \sqrt{1^2 + (\sqrt{2})^2 + (-1)^2} = \sqrt{4} = 2$$

$$\Rightarrow T(0) = \frac{r'(0)}{|r'(0)|} = \frac{1}{2} < 1, \sqrt{2}, -1 > = < \frac{1}{2}, \frac{\sqrt{2}}{2}, -\frac{1}{2} >$$

(b) Find parametric equations for the tangent line of the curve at the point with $t = 0$.

The tangent line pass through the point $r(0) = < 1, 0, 1 >$ and has the direction $r'(0) = < 1, \sqrt{2}, -1 >$

$$\Rightarrow \quad r(0) + t r'(0) = < 1, 0, 1 > + t < 1, \sqrt{2}, -1 > = < 1+t, \sqrt{2}t, 1-t >$$

$$\Rightarrow x = 1+t,\ y = \sqrt{2}t,\ z = 1-t.$$

(c) Find the length of the arc defined by $\mathbf{r}(t)$ with $0 \le t \le 1$.

$$|r'(t)| = \sqrt{(e^t)^2 + (\sqrt{2})^2 + (-e^{-t})^2} = \sqrt{e^{2t} + 2 + e^{-2t}} = \sqrt{(e^t + e^{-t})^2}$$

$$= e^t + e^{-t}$$

$$L = \int_0^1 |r'(t)| dt = \int_0^1 e^t + e^{-t} dt = [e^t - e^{-t}]_{t=0}^{t=1}$$

$$= [e - e^{-1}] - [1 - 1] = e - \frac{1}{e}$$